ATISHAY JAIN

Software Engineering | Full-Stack Development | Applied AI

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SUMMARY

Computer Engineering student with hands-on experience building production-oriented full-stack and AI systems using Python, FastAPI, React/Next.js, TypeScript, PostgreSQL, Docker, and modern LLM tooling. Built real-time web applications, secure backend workflows, offline AI pipelines, and CI/CD deployments, with a focus on reliability, performance, and iterative engineering.

EXPERIENCE

Def-Space Summer Intern - Bharat Space Education Research Centre (BSERC) **Jun 2026 - Aug 2026**

Def-Space Summer Internship Programme 2026 | 6+ weeks

- Completed a structured internship focused on Defence and Space technologies under the theme "Responsible Innovation for Inclusive Growth," gaining interdisciplinary exposure to engineering, research, and technology-led problem solving.
- Participated in technical learning activities designed to strengthen understanding of defence-space innovation and its role in India's scientific and technological development; awarded Certificate No. BSERC-DSI-2026-2601.

Finance & Documentation Manager - TEDxTIET **Aug 2025 - Feb 2026**

Patiala, Punjab, India

- Managed a ₹3.5 lakh event budget with structured allocation, real-time expense tracking, variance analysis, and sponsor documentation, improving financial visibility and operational control across teams.

PROJECTS

ReceiptSplit - UPI-Native Bill Coordination Platform **2026**

Next.js, TypeScript, FastAPI, PostgreSQL/Supabase, Docker, Realtime

- Built an end-to-end bill coordination flow covering receipt upload/OCR, editable line items, participant joining, item/equal/percentage/hybrid splits, shared items, partial quantities, taxes, discounts, and deterministic paise-level rounding.
- Designed FastAPI services and PostgreSQL/Supabase data models for authentication, real-time state, creator controls, settlement states, audit logging, abuse reporting, rate limiting, and server-generated UPI payment intents.
- Implemented a mobile-first Next.js interface and Dockerized development workflow; added migrations, test coverage, end-to-end smoke validation, and failure-state UX across receipt creation, claims, locking, payment, confirmation, and disputes.
- Hardened creator and participant flows with scoped permissions, safe status transitions, participant removal, settlement safeguards, and auditable events so the product remains understandable even when payments or claims change mid-session.

TIET-LOOP - Campus Event Platform **2025 - 2026**

React, Vite, Tailwind CSS, FastAPI, WebSockets, SQLite/PostgreSQL, Google Maps

- Developed a Progressive Web App centralizing campus events, RSVPs, real-time chat, friend interactions, carpool coordination, admin CRUD, calendar export, and map-based venue navigation in a single student-facing platform.
- Built FastAPI REST endpoints and WebSocket messaging with persistent storage and JWT-based flows, while a cosine-similarity recommendation engine personalized event discovery from user preferences and interaction signals.
- Integrated Google Maps for venue discovery and navigation, PWA support for mobile-first access, and event-management workflows that let organizers create, update, and manage campus listings from an admin interface.

Hybrid GenAI Transaction Categorizer **2025 - 2026**

Python, FastAPI, React, ONNX Runtime, Qwen2.5-7B GGUF, SQLite

- Built a privacy-first offline transaction classification pipeline combining merchant overrides, an ONNX-optimized transformer classifier, and a local LLM fallback for ambiguous cases and human-readable explanations.
- Added feedback-driven learning, a React dashboard, and FastAPI prediction APIs; optimized CPU-only local LLM inference from roughly 60–80 seconds to 5–13 seconds through prompt/context tuning and shared model execution.
- Structured the inference path so deterministic merchant rules handle known cases first, ONNX classification handles normal ambiguity, and a single shared Qwen2.5-7B instance is reserved for harder cases; learning is triggered explicitly through feedback rather than silently during prediction.

SKILLS

Languages: Python, C++, C, SQL, JavaScript/TypeScript, R

Full-Stack: React, Next.js, Vite, Tailwind CSS, FastAPI, Flask, Django, REST APIs, WebSockets

Data & Infra: PostgreSQL, Supabase, SQLite, MongoDB, Docker, Git, GitHub Actions, Linux, CI/CD

AI/ML: Transformers, ONNX Runtime, local LLMs, RAG, embeddings, Pandas, NumPy

EDUCATION & CERTIFICATIONS

B.Tech, Computer Engineering - Thapar Institute of Engineering and Technology **Aug 2023 - Jun 2027**

CGPA: 8 | Patiala, Punjab, India

Class XII (Science) - Sacred Heart Senior Secondary School, Ludhiana | 94.2%

Certifications: NVIDIA Foundations of Deep Learning; Duke University - Introduction to RAG; CI/CD Pipeline with Docker; Software Development fundamentals.

Achievement: Meta OpenEnv Hackathon 2026 Finalist - built Mahoraga, a self-improvement RL environment using Qwen2.5 and LoRA.